

More than just two peas in a pod: common amyloidogenic properties of tau and α -synuclein in neurodegenerative diseases

Virginia M-Y. Lee, Benoit I. Giasson and John Q. Trojanowski

The Center for Neurodegenerative Disease Research, Department of Pathology and Laboratory Medicine, and Institute on Aging, the University of Pennsylvania, Philadelphia, PA 19104-4283, USA

Intracytoplasmic filamentous aggregates, such as neurofibrillary tangles in Alzheimer's disease and Lewy bodies in Parkinson's disease, are composed of the proteins tau and α -synuclein, respectively. These pathological inclusions are linked directly to the etiology and mechanisms of disease in a wide spectrum of neurodegenerative disorders, termed 'tauopathies' and 'synucleinopathies'. Emerging evidence indicates that there is frequent overlap of the pathological and clinical features of patients with tauopathies and synucleinopathies, thereby re-enforcing the notion that these disorders might be linked mechanistically. Indeed, several lines of investigation suggest that tau and α -synuclein might constitute a unique class of unstructured proteins that assemble predominantly into homopolymeric (rather than heteropolymeric) fibrils, which deposit mainly in separate amyloid inclusions, but occasionally deposit together. Thus, the ability of tau and α -synuclein to affect each other directly or indirectly might contribute to the overlap in the clinical and pathological features of tauopathies and synucleinopathies.

Alzheimer's disease (AD) and Parkinson's disease (PD) are the most common neurodegenerative disorders associated with cognitive and motor impairments, respectively. Both diseases are characterized neuropathologically by the presence of filamentous inclusions in affected neurons in selectively vulnerable regions of the CNS. The loss of neurons from brain regions that subserve specific cognitive or motor functions is thought to be responsible for the distinct clinical phenotypes classically described for AD and PD. The filamentous neuronal inclusions of AD, known as neurofibrillary tangles (NFTs), are composed of pathologically altered forms of the microtubule (MT)-associated protein tau [1]. Similarly, the signature inclusions of PD, termed Lewy bodies (LBs), are comprised of fibrils assembled from the presynaptic protein α -synuclein [2]. Neurofibrillary tau lesions are not restricted to AD; they are also the defining neuropathological hallmarks of other neurodegenerative diseases, such as

frontotemporal dementias (FTDs) including Pick's disease, progressive supranuclear palsy (PSP) and corticobasal degeneration (CBD) – conditions that are collectively referred to as tauopathies [1]. Furthermore, although α -synuclein-containing LBs are the defining characteristics of PD, several other neurodegenerative diseases show abundant α -synuclein inclusions. These disorders, known as synucleinopathies, include dementia with Lewy body (DLB), rapid-eye-movement sleep disorders, progressive autonomic failure and multiple system atrophy (MSA) [2].

Autosomal-dominantly inherited *tau* and *α -synuclein* mutations can be pathogenic in FTD with parkinsonism linked to chromosome 17 (FTDP-17) [1,3,4] and in familial PD [2,5], respectively. This suggests that both tau and α -synuclein lesions can directly cause neurodegenerative diseases. Indeed, compelling evidence accumulated over the past five years supports the notion that intracellular tau and α -synuclein pathologies cause disease both by a loss of normal function and by a gain of one or more toxic functions. In particular, the aggregation of tau and α -synuclein into fibrillar amyloid deposits can, in essence, lead to the sequestration of these proteins and prevent them from performing their normal functions, thereby 'taking them out of action' and leading to a loss of function [1–3]. Conversely, strong evidence indicates that intracytoplasmic amyloid inclusions composed of filamentous forms of α -synuclein and tau might be directly cytotoxic because these inclusions can impair or obstruct normal cellular trafficking, thereby disrupting cell morphology and homeostasis, and can trap other cellular components, stopping them from performing their normal functions [1,3,6]. However, the neurotoxic mechanisms of LBs and NFTs remain controversial and opinions even differ as to whether or not these lesions are protective or deleterious to the neurons that harbor them. Although additional studies are needed to resolve the neurotoxic mechanisms, the divergent opinions on the effects of lesions might not be mutually exclusive. From a dynamic perspective of disease mechanisms, one might imagine that inclusion formation could reflect the activation of pathways that are neuroprotective early in the disease but that later have

Corresponding author: Virginia M-Y. Lee (vmylee@mail.med.upenn.edu).

deleterious consequences, when the inclusions become so voluminous that they compromise the function and viability of affected cells. These uncertainties notwithstanding, we compare the biochemical and the neuropathological properties of tau and α -synuclein in this review. Because of the striking similarities between these proteins in tauopathies and synucleinopathies, we propose that tau and α -synuclein define a new class of amyloid deposits in aging-related neurodegenerative diseases.

Tau proteins in tauopathies

Tau proteins are MT-binding proteins that stabilize and promote MT polymerization in neuronal perikarya and processes [1,7]. In the adult human brain, six isoforms ranging between 352 and 441 amino acids in length are produced as a result of alternative mRNA splicing from a single gene on chromosome 17 [1,7]. Tau protein monomers can polymerize to form fibrils known as paired helical filaments (PHFs), which are the building blocks of the NFTs, neuropil threads and dystrophic neurites that accumulate in the brains of patients with AD, Pick's disease, PSP and CBD. Although tau is predominantly a neuronal protein, tau pathologies can accumulate in oligodendrocytes and astrocytes, in addition to neurons, in diverse neurodegenerative disorders [1,7]. The identification of *tau* mutations in families with FTDP-17 has provided unequivocal evidence that alterations in tau properties can cause disease [4], and *in vitro* studies have demonstrated that these mutations can result in reduced binding of MTs by tau and increased tendency of tau to polymerize into fibrils that resemble those in pathological inclusions [8–11]. Studies of transgenic mouse lines expressing FTDP-17 mutant tau protein also support the notion that the increased propensity of tau to form filamentous inclusions can lead to disruption of cellular homeostasis, cell death and disease [12–14].

α -Synuclein and synucleinopathies

Native α -synuclein is a small (140 amino acid) protein that is predominantly expressed in CNS neurons, where it is concentrated at presynaptic terminals [15–18]. The function of α -synuclein is still not fully characterized but several studies suggest that it is involved in modulating synaptic transmission, regulating the localization of vesicles in synapses and participating in neuronal plasticity [17,19,20].

Significantly, α -synuclein was directly implicated in neurodegenerative disease when the Ala53Thr mutation (resulting from a G to A transition at position 209 of the *α -synuclein* gene) was identified in a large Italian family (the Contursi kindred) and in three small, possibly related, Greek families with autosomal-dominant PD [5]. Shortly following the report of the Ala53Thr mutation, a series of studies demonstrated that α -synuclein assembles into fibrils. Furthermore, these fibrils were shown to be the building blocks of pathological amyloid inclusions known as LBs and Lewy neurites, which are characteristic lesions of many neurodegenerative disorders including PD, DLB and the LB variant of AD (LBVAD). Similar fibrils form glial cytoplasmic inclusions that are diagnostic of MSA [2,21–27]. α -Synuclein pathology is also present in a

subset of patients afflicted by several other neurodegenerative disorders (e.g. progressive autonomic failure and neurodegeneration with brain iron accumulation, type 1). Together, these findings lead to the classification of these diseases as synucleinopathies [2].

Similarities of tau and α -synuclein in tauopathies and synucleinopathies

In addition to the fact that tau and α -synuclein pathological inclusions are found in a spectrum of neurodegenerative disorders, the biological and biophysical properties of normal and pathological tau and α -synuclein suggest that they could define a new class of proteins that form brain amyloid deposits. For example, both tau and α -synuclein are abundant neuronal proteins that have a high content of charged amino acid residues and are highly soluble in aqueous buffers [16,28–30]. Moreover, they have long half-lives *in vivo* and can exist as 'natively unfolded', with no apparent defined structure, which allows them to be heat-stable [28,29]. They also contain stretches of hydrophobic residues that are involved in forming the core of the assembled fibrils. Further, both proteins are phosphorylated, and these modifications have been implicated in facilitating their conversion into fibrils [31,32]. α -Synuclein and tau fibrils, either assembled *in vitro* from recombinant protein or as observed from pathological inclusions, share common properties with amyloid: they form 10–15 nm diameter filaments that are detected by amyloid stains such as Congo Red and Thioflavin S [1,2,33–36].

Moreover, tau and α -synuclein in pathological inclusions share common abnormal biochemical properties such as aberrant hyperphosphorylation, ubiquitination and nitration [1,2,37–40]. Furthermore, both fibrillar tau and α -synuclein lesions often sequester other cytoskeletal proteins, such as neurofilaments, and they can be detected as intracytoplasmic inclusions in neurons and glia, as neuro-axonal spheroids and as 'threads' (or Lewy neurites) in neuronal and glial processes. Of particular interest is the formation of glial tau and α -synuclein inclusions in MSA [41] (Figure 1) and of glial tau pathologies in Pick's disease, PSP and CBD [1] because both proteins are primarily neuronal and are normally expressed only at very low levels in glial cells. Thus, the abnormal accumulation of tau and α -synuclein into insoluble aggregates in glial cells in disorders such as MSA also suggests a common pathogenic mechanism for the formation of tau and α -synuclein amyloid deposits.

Co-occurrence of tau and α -synuclein amyloidogenic inclusions

Indeed, in addition to MSA, both tau and α -synuclein inclusions have been reported in several neurodegenerative diseases and sometimes both lesions even occur in the same cells (Figure 1), with the tau and α -synuclein fibrils occasionally appearing to be intertwined [42–45]. For example, α -synuclein and tau pathologies have frequently been observed in brains from patients with familial AD (due to mutations in the genes encoding amyloid precursor protein, presenilin-1 and presenilin-2) [46],

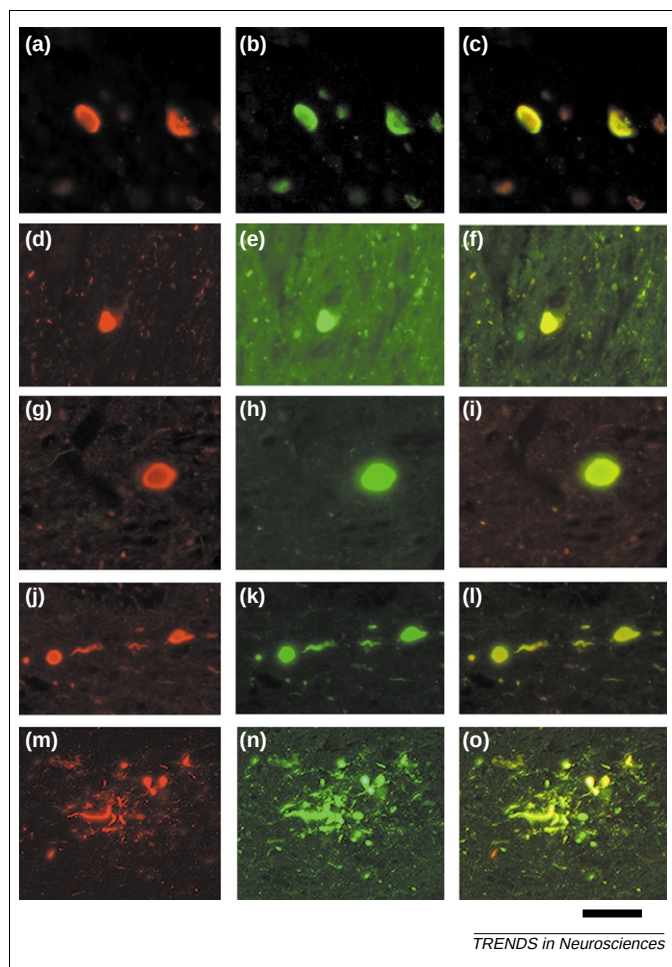


Figure 1. Colocalization of α -synuclein and tau in some pathological inclusions. (a–c) Glial cytoplasmic inclusions in the cerebellar white matter of a patient with multiple system atrophy. (d–f) A Pick body in the hippocampus of a patient with Pick's disease. (g–i) A Lewy body and (j–l) spheroids in the substantia nigra pars compacta of a patient with Parkinson's disease. (m–o) Dystrophic neurites within a senile plaque of a patient with the Lewy Body variant of Alzheimer's disease. Pathological lesions were stained by immunofluorescence with a rabbit anti- α -synuclein antibody (left panels; red) and a mouse anti-tau antibody (middle panels; green). The overlays of both stains are shown in the right panels. Scale bar, 20 μ m for all panels.

Down's syndrome [44] and parkinsonism-dementia complex of Guam [47,48].

A direct link between the aggregation of tau and α -synuclein was demonstrated in the pathological re-evaluation of members of the Contursi kindred with the Ala53Thr α -synuclein mutation. Remarkably, post-mortem examination of two siblings from this pedigree revealed widespread α -synuclein pathological inclusions, as well as tau inclusions [49,50]. Although inclusions formed by α -synuclein aggregation alone occur in these patients, many contain both α -synuclein and tau, and in some brain regions the disease inclusions are comprised solely of tau [49]. These findings established that a pathogenic mutation in α -synuclein that increases the propensity of α -synuclein to fibrillize [51,52] also promotes the formation of filamentous tau inclusions.

Interaction of tau and α -synuclein in promoting amyloidosis

To gain insights into how alterations in α -synuclein can cause the formation of tau lesions, we adopted an *in vitro*

approach to study the possibilities that α -synuclein and tau might interact and that this association could have an effect on polymerization of the proteins into fibrils. Previous studies have shown that α -synuclein readily self-polymerizes *in vitro* [51,52], whereas tau requires cofactors (e.g. polyanions such as glycosaminoglycans or nucleic acids) to fibrillize [53,54]. Significantly, we showed that α -synuclein can substitute for polyanions in that it can initiate the polymerization of tau *in vitro* [49]. Moreover, once tau fibrillization is initiated, tau and α -synuclein synergistically promote and propagate the polymerization of each other into amyloid fibrils: low concentrations of α -synuclein do not readily fibrillize in the absence of tau but the presence of tau significantly increases assembly of α -synuclein fibrils [49]. Interestingly, despite the biochemical and biophysical similarity between tau and α -synuclein, each of these proteins preferentially formed homopolymers rather than heteropolymeric filaments *in vitro* once fibrillization was initiated by α -synuclein [49]. These findings suggest that, rather than facilitating the complete assembly of tau filaments, α -synuclein might enhance a crucial rate-limiting step in the initiation and/or early stages of tau fibrillization (Figure 2).

The ability of α -synuclein to induce the formation of tau inclusions and the synergistic aggregation of both proteins was further supported by studies of transgenic mice expressing Ala53Thr mutant human α -synuclein. Tau inclusions were observed in ~50% of these mice, concomitant with the development of α -synuclein filamentous inclusions that lead to severe motor dysfunction [49]. Furthermore, although neither α -synuclein nor tau inclusions were present in transgenic mice expressing either α -synuclein or mutant Pro301Leu tau alone in oligodendrocytes, filamentous amyloid inclusions composed of both proteins were observed in bigenic mice expressing tau and α -synuclein [49].

These results clearly demonstrate that α -synuclein can participate in the formation of tau fibrils and that amyloid fibril formation by both proteins can share similar mechanisms. However, the precise molecular mechanisms leading to the enhancement or synergies of the fibrillization remain to be fully elucidated. Nevertheless, the amyloidogenic properties of α -synuclein are crucial to the interaction between α -synuclein and tau, as demonstrated by the inability of β -synuclein and an artificial α -synuclein (Δ 71–82) to induce tau fibrillogenesis [49]. These two synuclein proteins are similar to α -synuclein in size and amino acid sequences, but they cannot form amyloid owing to the lack of hydrophobic amino acid stretches in their middle regions [55]. Furthermore, there seems to be some specificity in the interactions of amyloidogenic proteins because the β -amyloid peptide is not able to induce tau filament formation [49]. The formation of amyloid fibrils in the presence of both α -synuclein and tau appears to involve a two-step mechanism (initiation followed by propagation). The initiation step probably involves the formation of amyloidogenic α -synuclein 'seeds' that can prime tau to acquire a conformation that is polymerization-permissive (Figure 2). In this amyloidogenic conformation, tau and α -synuclein might be able to alter the structural properties of monomers of both proteins. However, the

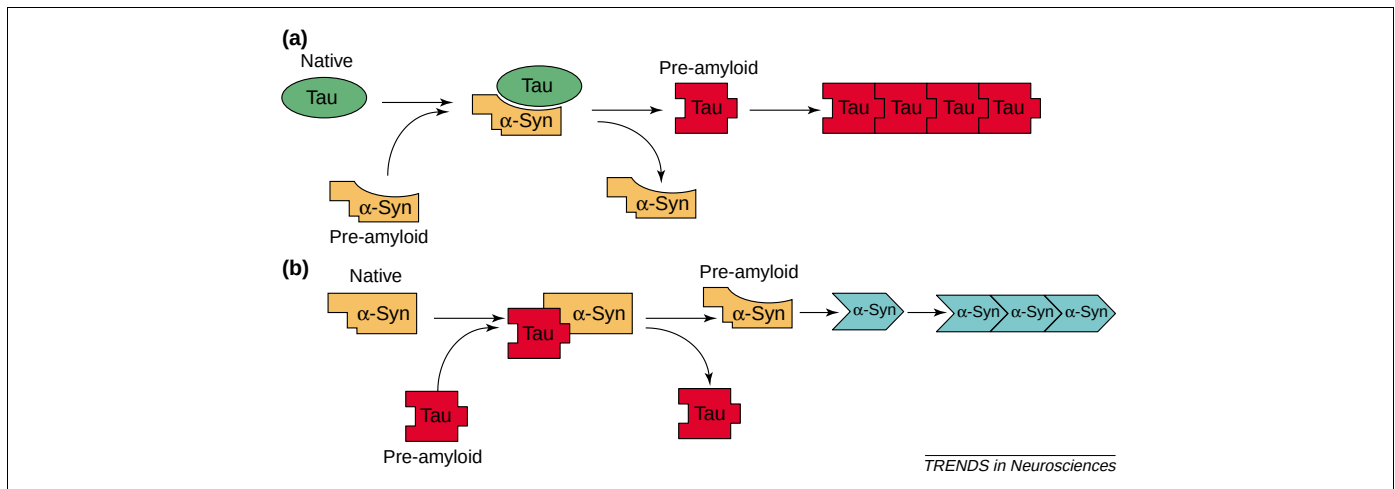


Figure 2. Putative interactions between α -synuclein and tau that could synergistically promote fibril formation. **(a)** Native tau (green) can interact with pre-amyloidogenic species ('seeds') of α -synuclein (orange) resulting in the conversion of tau into a pre-amyloidogenic conformation (red). These tau molecules can then polymerize into fibrils. **(b)** Native α -synuclein (orange) can interact with pre-amyloidogenic species of tau (red). This interaction facilitates the transformation of α -synuclein into pre-amyloid precursors (blue). These precursors could undergo further conformation changes before they elongate as homopolymers.

elongation of small oligomers into higher-ordered proteins complexes seems to be specific for each protein, as they preferentially form filamentous homopolymers.

These studies unequivocally demonstrate that α -synuclein and tau can associate and promote the fibrillogenesis of each other *in vitro* and in transgenic mouse models. However, there are other unique biological properties of these molecules that must be considered in order to determine if and how these proteins interact in human CNS neurons. For example, in healthy adult neurons, both proteins are localized mainly to different neuronal compartments – tau proteins are predominantly bound to MTs in axons and cell bodies, whereas α -synuclein is localized to vesicles at the presynapse (Figure 3) [1,2,7,19]. Therefore, it is likely that for tau– α -synuclein interaction to occur, changes in the cellular physiology of each protein must occur first; these changes might reduce the ability of α -synuclein and/or tau to perform their normal functions, resulting in their aberrant colocalization. Although α -synuclein is concentrated in the axonal terminal in adult neurons, it is also present at low levels in axonal processes, while it is transported to the terminals. Aberrant trafficking of α -synuclein might lead to its accumulation in cell bodies and processes. Furthermore, the hyperphosphorylation of tau that occurs in disease states could increase the pool of MT-dissociated tau, which could then interact with α -synuclein. The rather high concentrations and long half-lives of tau and α -synuclein in neurons make it is plausible that, under pathological conditions, both proteins might be able to reach threshold concentrations that surpass their normal physiological levels and, hence, lead to the formation of filamentous inclusions.

Closing remarks

Current data indicate that α -synuclein can be a pathological initiator of tau amyloid formation [49]. However, it is likely to be only one of several cofactors or 'pathological chaperones' that can induce the fibrillization of tau. The notion that there may be multiple initiators of tau

polymerization is consistent with the findings that α -synuclein inclusions are detected in only a subset of cells with prominent tau pathology in diseased brains. Moreover, α -synuclein pathology has not been reported in FTDP-17 patients with tau mutations [1,3,56,57], although it does occur in tau lesions of Guam tauopathies that appear to be familial. Nevertheless, it is tempting

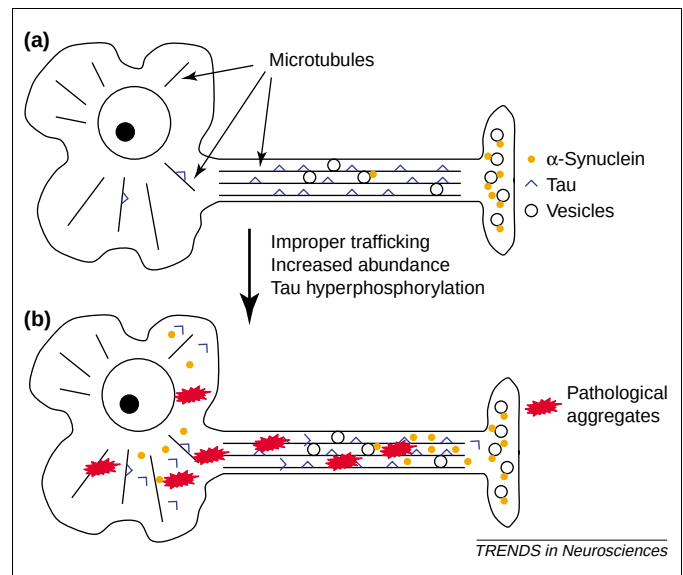


Figure 3. The normal distributions of α -synuclein and tau, and the pathological changes that might cause them to interact pathologically in diseased neurons. **(a)** In normal neurons, α -synuclein is concentrated at axonal terminals where it is in close proximity to synaptic vesicles. Tau predominantly interacts with microtubules in perikarya and axons. In adult CNS neurons, α -synuclein and tau are unlikely to interact with each other, except when low levels of α -synuclein are expressed in the cell body before being transported to synaptic terminals. **(b)** Under pathological conditions, aberrant cellular trafficking or axonal transport can lead to accumulation of α -synuclein in cell bodies and axons. There might also be an increased abundance of α -synuclein or tau owing to increased expression or reduced turnover, which could lead to focal accumulations of these proteins in cellular compartments where they are not normally found together. Further, the hyperphosphorylation of tau that occurs in diseased neurons reduces the affinity of tau for microtubules. This could lead to an increase in free cytoplasmic tau, which might then associate with α -synuclein. These and other mechanisms could promote the abnormal interaction of tau and α -synuclein, thereby increasing the likelihood that they can facilitate the fibrillogenesis of each other and lead to inclusion formation.

to speculate that α -synuclein could be able to alter the conformation or solubility of tau in brains without leaving morphological 'traces' of obvious α -synuclein pathology. Consistent with this notion are the facts that only minute amounts of amyloidogenic α -synuclein might be required to initiate tau fibrillogenesis and that the small amounts of the α -synuclein 'seeds' required for promoting fibrillization of tau might go undetected by current methods. Alternatively, small amounts of amyloidogenic α -synuclein might be degraded following their initiation of tau polymerization. Although this concept is provocative, it is a hypothesis that can be tested in future studies designed to improve current understanding of the interactions of amyloidogenic brain polypeptides in diverse neurodegenerative diseases. The recent studies summarized here provide novel insights into common mechanisms underlying the formation of pathological inclusions in AD, PD, LBVAD and related disorders. Importantly, these findings also provide a strong rationale for pursuing efforts to identify therapeutic agents that could directly or indirectly inhibit the formation of amyloid, and thereby show efficacy for intervening in several brain amyloidosis disorders.

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